

Global Attention-Based CNN for Interpretable Gender Classification in Palm Vein Biometrics

Sorn Sooksatra and Sitapa Watcharapinchai

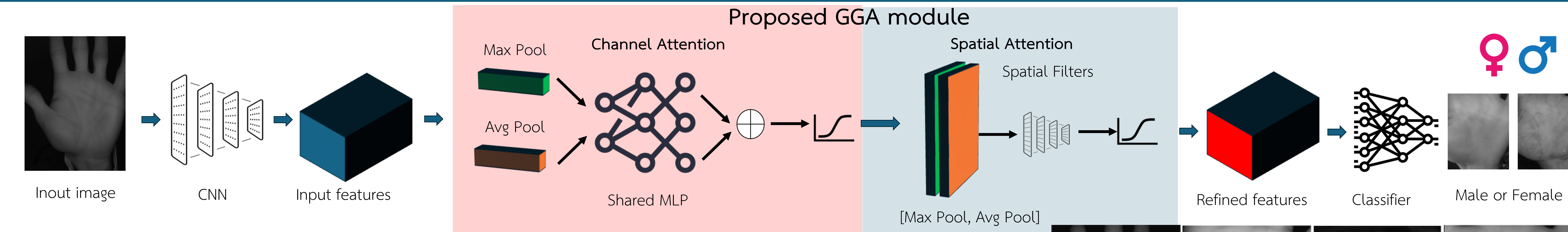
National Electronic and Computer Technology Center, National Science and Technology Development Agency, Khlong Luang, Pathum Thani, 12120, Thailand

Abstract

Our **GGA-CNN** framework addresses gender classification bias in palm vein biometrics by introducing a **Global Gated Attention (GGA)** module to prioritize internal vascular geometry. Evaluated on the gender-labeled **VERA dataset**, the proposed method achieves baseline CNNs accuracy of **94.39%**. By refining features at the terminal bottleneck, the GGA module successfully localizes the **Region of Interest (ROI)** to the palm center, significantly improving classification precision and mitigating performance disparities for female subjects.

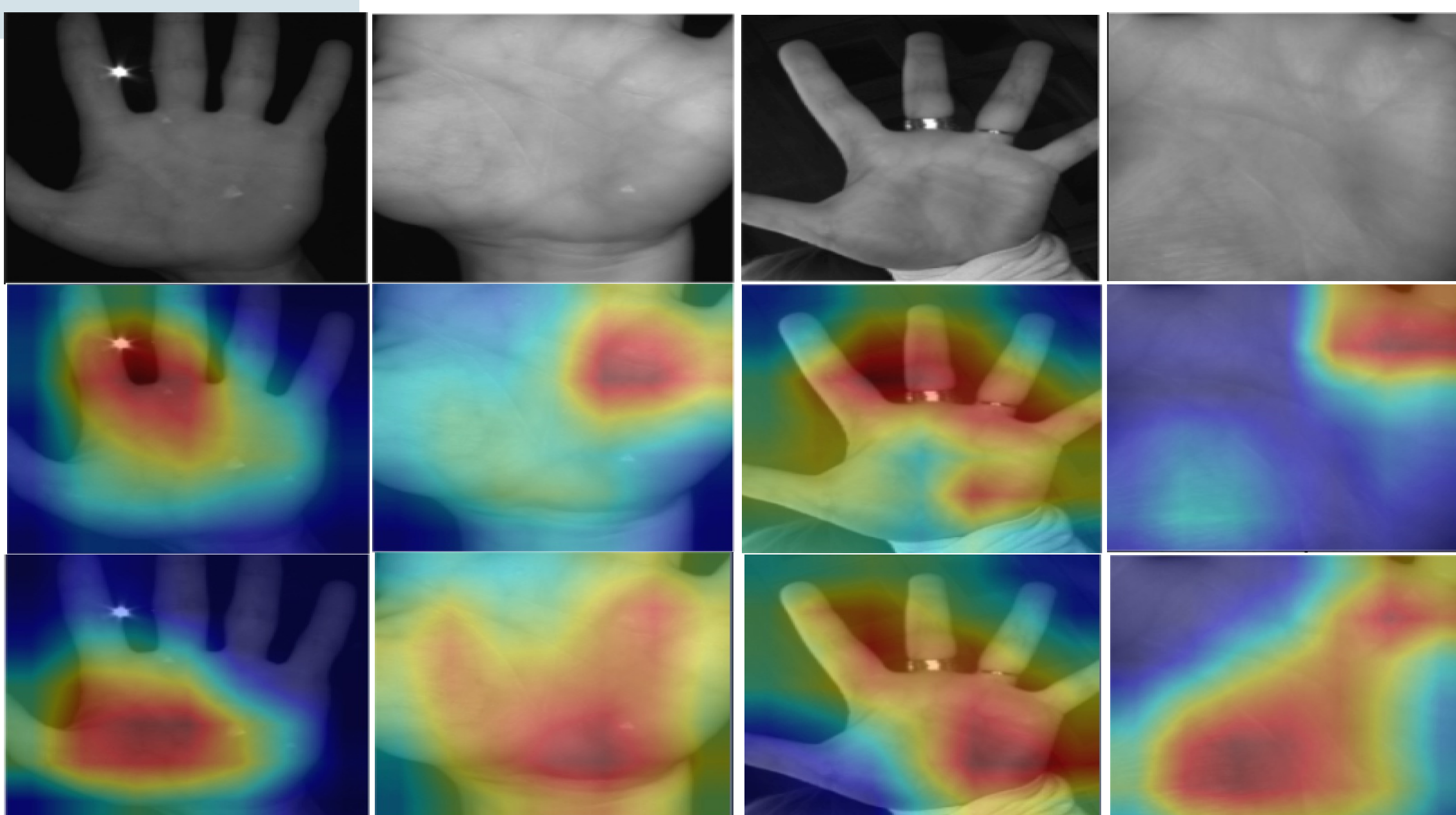
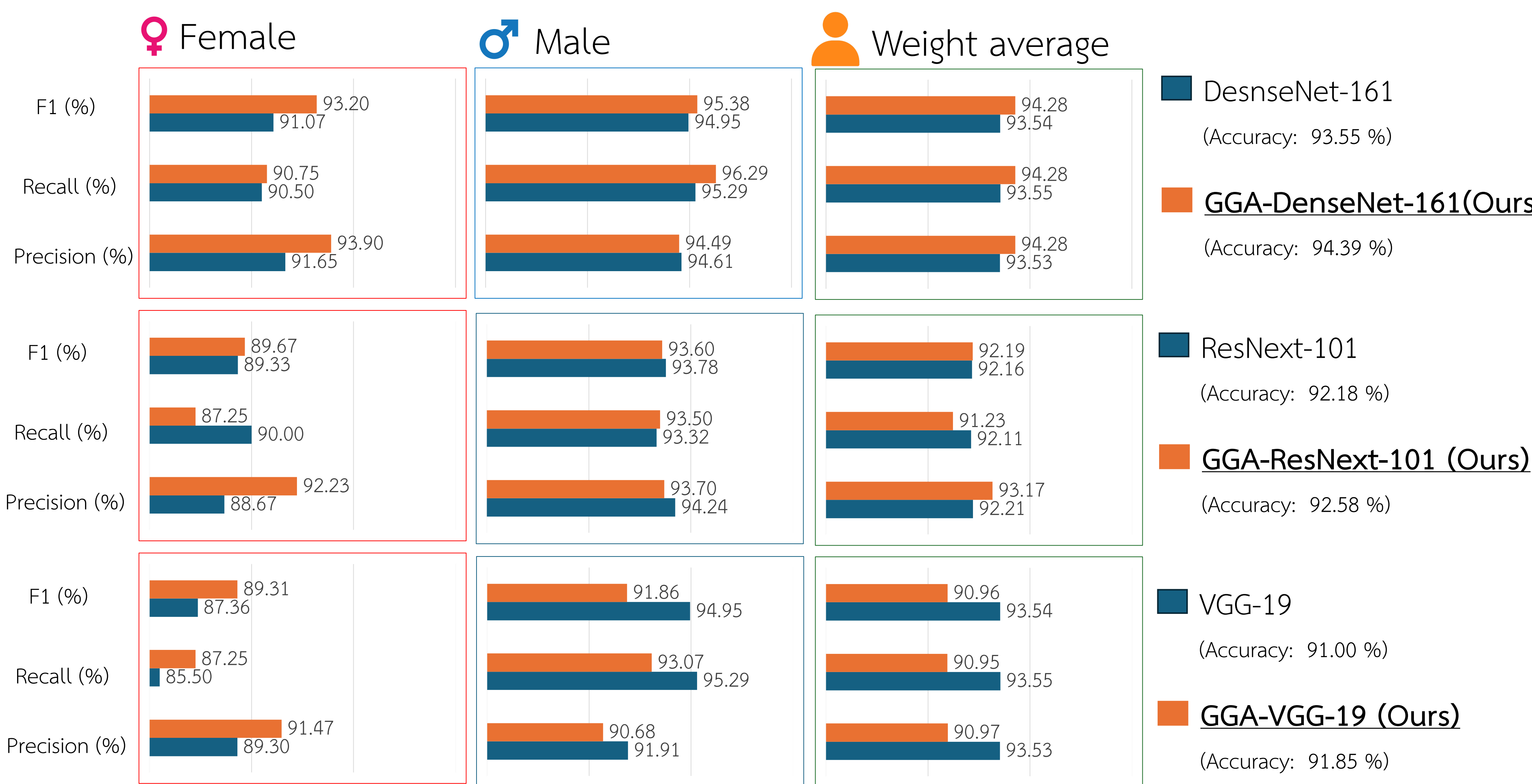
Background & motivation

- The Problem:** While palm vein recognition is highly reliable for identity, gender classification remains a challenge due to the subtle physiological variations in vascular distribution.
- The Gap:** Standard CNN backbones often suffer from a lack of spatial focus, frequently attending to irrelevant peripheral hand contours rather than the central palmar arch.
- Unique Data Source:** This study utilizes the **VERA Palm Vein Database** [1]. It is one of the few publicly available biometric repositories that provides the **explicit gender metadata**.
- Baseline Inspiration:** Following the demographic analysis by [2], we aim to amplify the specific vascular diameter and distribution patterns that distinguish male and female subjects.



Methodology

- Proposed Innovation:** We introduce the **Global Gated Attention (GGA)** module, a "plug-and-play" architecture-agnostic extension.
- Concept Evolution:** While our gating mechanics draw foundational inspiration from the **CBAM** dual-attention concept [3], we re-engineer them into a single, macro-level filter.
- Strategic Placement:** Unlike traditional block-wise attention, the GGA is placed exclusively at the **terminal feature bottleneck** to refine high-level semantic features right before classification.



Grad-CAM visualizations comparing feature localization from input images (top) with DenseNet-161 (center) and GGA-DenseNet-161 (bottom)

Results & Discussion

- Peak Accuracy:** The accuracy of **GGA-DenseNet-161** configuration outperformed all other models. Our method significantly improved the F1-score for female subjects, addressing a common performance imbalance in gender classification.
- Interpretability (Grad-CAM):** Visualizations confirm that the GGA module successfully re-centers the network's focus onto the **central palm ROI**, effectively attenuating noise from skin texture.
- Efficiency:** By applying attention only at the global stage, we achieve superior feature refinement without the massive computational overhead of repetitive, block-level attention modules.

Conclusion: The proposed **Global Gated Attention (GGA)** module provides a robust, plug-and-play extension that successfully localizes gender-discriminative vascular features in the palm center. Evaluation on the **VERA database** shows our GGA-DenseNet-161 configuration outperforms a traditional methods. This framework proves that terminal, macro-level attention significantly improves biometric classification precision.

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Reference

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